Question	Answer	Mark	Mark scheme	Additional guidance
20	Proof	M1	for $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ **or** $\overrightarrow{BA} = \mathbf{a} - \mathbf{b}$ **or** $\overrightarrow{AP} = \frac{4}{5}(\mathbf{b} - \mathbf{a})$ **or** $\overrightarrow{BP} = \frac{1}{5}(\mathbf{a} - \mathbf{b})$	Vectors could be written on the diagram
		M1	for $\overrightarrow{OP} = \mathbf{a} + \frac{4}{5}(\mathbf{b} - \mathbf{a})$ **or** $\overrightarrow{OP} = \mathbf{b} + \frac{1}{5}(\mathbf{a} - \mathbf{b})$ **or** $\overrightarrow{PC} = \frac{4}{5}(\mathbf{a} - \mathbf{b}) + 4\mathbf{b}$ **or** $\overrightarrow{OC} = \mathbf{a} + 4\mathbf{b}$	
		M1	for correct expressions, in terms of **a** and **b**, for two of $\overrightarrow{OP}$, $\overrightarrow{PC}$, $\overrightarrow{OC}$ eg $\overrightarrow{OP} = \mathbf{a} + \frac{4}{5}(\mathbf{b} - \mathbf{a})$ **or** $\overrightarrow{OP} = \mathbf{b} + \frac{1}{5}(\mathbf{a} - \mathbf{b})$ **or** $\overrightarrow{PC} = \frac{4}{5}(\mathbf{a} - \mathbf{b}) + 4\mathbf{b}$ **or** $\overrightarrow{OC} = \mathbf{a} + 4\mathbf{b}$	
		M1	for correct simplified expressions, in terms of **a** and **b**, to enable comparison of two of $\overrightarrow{OP}$, $\overrightarrow{PC}$, $\overrightarrow{OC}$ eg two of $\overrightarrow{OP} = \frac{1}{5}\mathbf{a} + \frac{4}{5}\mathbf{b}$ oe, $\overrightarrow{PC} = \frac{4}{5}\mathbf{a} + \frac{16}{5}\mathbf{b}$ oe, $\overrightarrow{OC} = \mathbf{a} + 4\mathbf{b}$	
		C1	for a complete proof with statement, eg $\overrightarrow{PC} = 4\overrightarrow{OP}$ so the vectors are parallel and have point *P* in common	